

15 Which statement about a simple harmonic oscillator is correct?

- A** The maximum kinetic energy of the oscillator is proportional to the frequency of its motion.
- B** The potential energy of the oscillator is minimum when the oscillator is momentarily at rest.
- C** The kinetic energy of the oscillator is zero when the oscillator is at the equilibrium position.
- D** When the kinetic energy of the oscillator is equal to its potential energy, the oscillator is neither at the rest position nor at the maximum displacement positions.

16 A body performs simple harmonic motion with a period of 0.063 s . The maximum speed